
FINAL PROJECT REPORT / 2026

ICT 1209 - Web Technologies
Department of ICT
Rajarata University of Sri Lanka

Falcon Perfumes

Phase 3 - PHP & MySQL Integration

THEME

E-commerce / Online Perfume Store

COURSE

ICT 1209 - Web Technologies

REPORT DATE

27 September 2026

GROUP MEMBERS

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INDEX NUMBERS

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GitHub Repository : <https://github.com/mubarakmr209-rgb/Falcon-Perfumes>

01 / Project overview

A database-backed fragrance shopping experience

Falcon Perfumes is a perfume retail website that helps customers explore fragrances, inspect product photos and notes, and place cash-on-delivery orders. The target audience is fragrance shoppers looking for a clear product catalogue and store staff who need a simple way to maintain stock and orders.

The starting point was four supplied HTML pages: Home, New Arrivals, Brands and Contact. The submission converts their page roles into shared PHP templates, recreates missing CSS and JavaScript assets, and retains the navy-and-gold visual identity. Generic stock photos were replaced by product-matched images.

Requirements and delivered features

Requirement	Implementation
Theme database	falcon_perfumes; ten related MySQL/MariaDB tables
Users table	Unique username/email, bcrypt password and created_at
Contact/messages	Validated contact form, database storage and confirmation
Registration and login	PHP handlers, password verification and session regeneration
Protected dashboard	Customer order history; separate protected admin area
Theme-specific data	Products, photos, brands, categories, orders and order items
PHP form processing	State-changing requests use POST, CSRF checks and PDO
Frontend validation	HTML validation plus JavaScript checks; PHP remains authoritative
Admin management	Product CRUD, gallery, stock, orders, customers and messages
Submission documents	Source ZIP, SQL export, report PDF, setup and GitHub instructions

The application has no external CSS/JavaScript CDN dependency and uses bundled images. It runs locally without an internet connection after XAMPP and the database are set up.

02 / System design

Database schema and relationships

All application tables use InnoDB and utf8mb4. Foreign keys maintain relationships; DECIMAL fields store monetary values. Product and customer records are retained where historic orders depend on them.

Table	Main fields / purpose
users	id PK, username UNIQUE, email UNIQUE, password, is_active, created_at
admins	id PK, username UNIQUE, password, must_change_password, created_at
messages	id PK, name, email, message, is_read, created_at
brands / categories	id PK, name UNIQUE; catalogue classification
products	id PK, brand_id FK, category_id FK, name, concentration, size_ml, description, top/heart/base notes, price, stock, is_sold_out, is_active, created_at
product_images	id PK, product_id FK, path, sort_order
orders	id PK, user_id FK, customer_name, email, phone, address, subtotal, shipping, total, status, payment_method, checkout_token UNIQUE, created_at
order_items	id PK, order_id FK, product_id FK, product_name snapshot, unit_price snapshot, quantity
login_attempts	id PK, hashed_attempt_key, created_at; basic sign-in throttling

Relationship map

Parent	Relationship	Child
users	1 to many	orders
orders	1 to many	order_items
products	1 to many	order_items and product_images
brands / categories	1 to many	products

Admins are independent of customer accounts. Messages can be submitted without signing in. Product deletion archives the listing instead of removing rows needed by order history. Order items preserve the purchased name and price even after catalogue edits.

03 / System design

Folders, routes and request processing

Path	Responsibility
css/ and js/	Responsive styling, mobile menu, gallery and client validation
images/products/	Product-matched local photos
images/uploads/	Validated uploads with random filenames and execution restrictions
includes/db.php	PDO connection and prepared-query helper
includes/functions.php	Sessions, CSRF, account guards, shared cart and product helpers
includes/header.php / footer.php	Shared layout for store and staff pages
auth/	register.php, login.php and logout.php
admin/	Protected dashboard, products, gallery, taxonomy, orders, customers and inbox
index.php / new-arrivals.php	Store landing page and searchable, filterable catalogue
product.php / brands.php	Product detail gallery and brand browsing
cart.php / checkout.php	Session bag and transactional order placement
contact.php / dashboard.php	Contact submission and customer order history
database.sql / README.md	Importable clean database export and setup guide
docs/ / tests/ / tools/	Report source, screenshots, references, smoke checks and local admin reset

How a request is processed

A PHP page loads the shared functions and PDO configuration. Protected routes validate the relevant customer or admin session. On POST, the handler checks the CSRF token and validates all submitted values before using a prepared SQL statement. Successful writes redirect to a fresh GET request, which displays a confirmation message and current database values.

The product page is reached by selecting a product card image or name. Its ID selects the database record and gallery; invalid or archived products return HTTP 404. There is no separate Product Page item in the navigation.

04 / Implementation

Customer authentication and account history

Registration accepts a username, email and password with confirmation. PHP checks input length, username format, email validity, uniqueness and password matching. Passwords use `password_hash(..., PASSWORD_BCRYPT)`; login uses `password_verify()`. Password input is limited to 10-72 bytes to avoid bcrypt truncation.

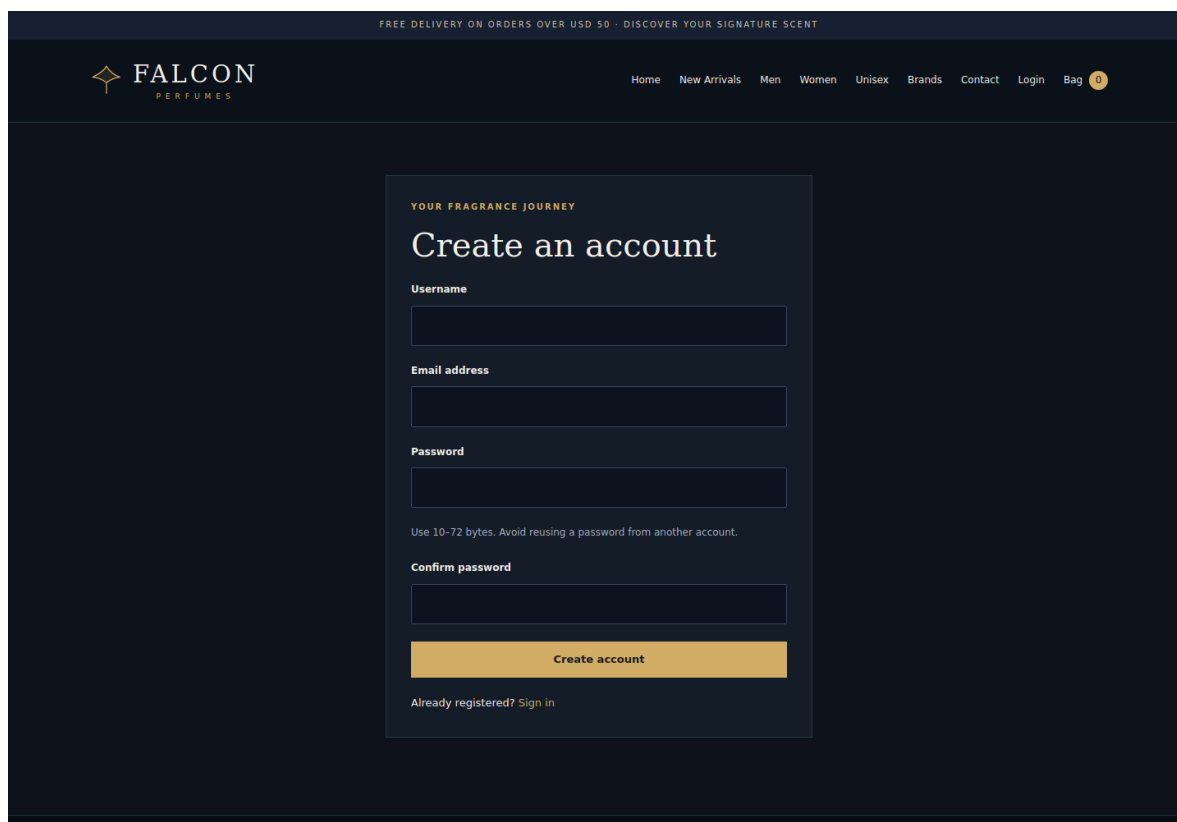


Figure 1. Registration page rendered by the PHP application.

After valid login, `session_regenerate_id(true)` replaces the session ID before the customer reaches `dashboard.php`. Logout accepts a CSRF-protected POST and destroys the session and cookie. Five failed sign-ins for the same identity/IP within 15 minutes temporarily block further attempts.

The customer dashboard queries only orders belonging to the current account. Disabling a customer in the admin panel prevents future login and removes protected-page access on the next request. Admins use a separate table, session name and cookie path; customer registration cannot grant staff privileges.

05 / Implementation

Correct products, descriptions and image galleries

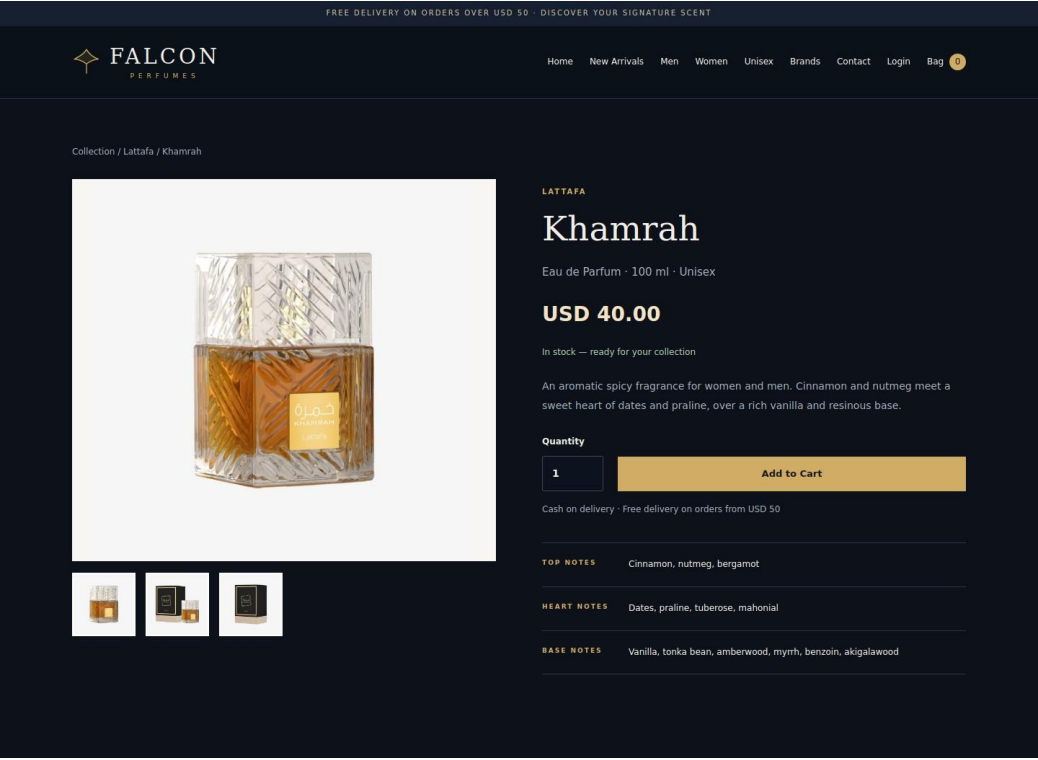


Figure 2. Khamrah detail page, reached from its product card; gallery and note stages are database-backed.

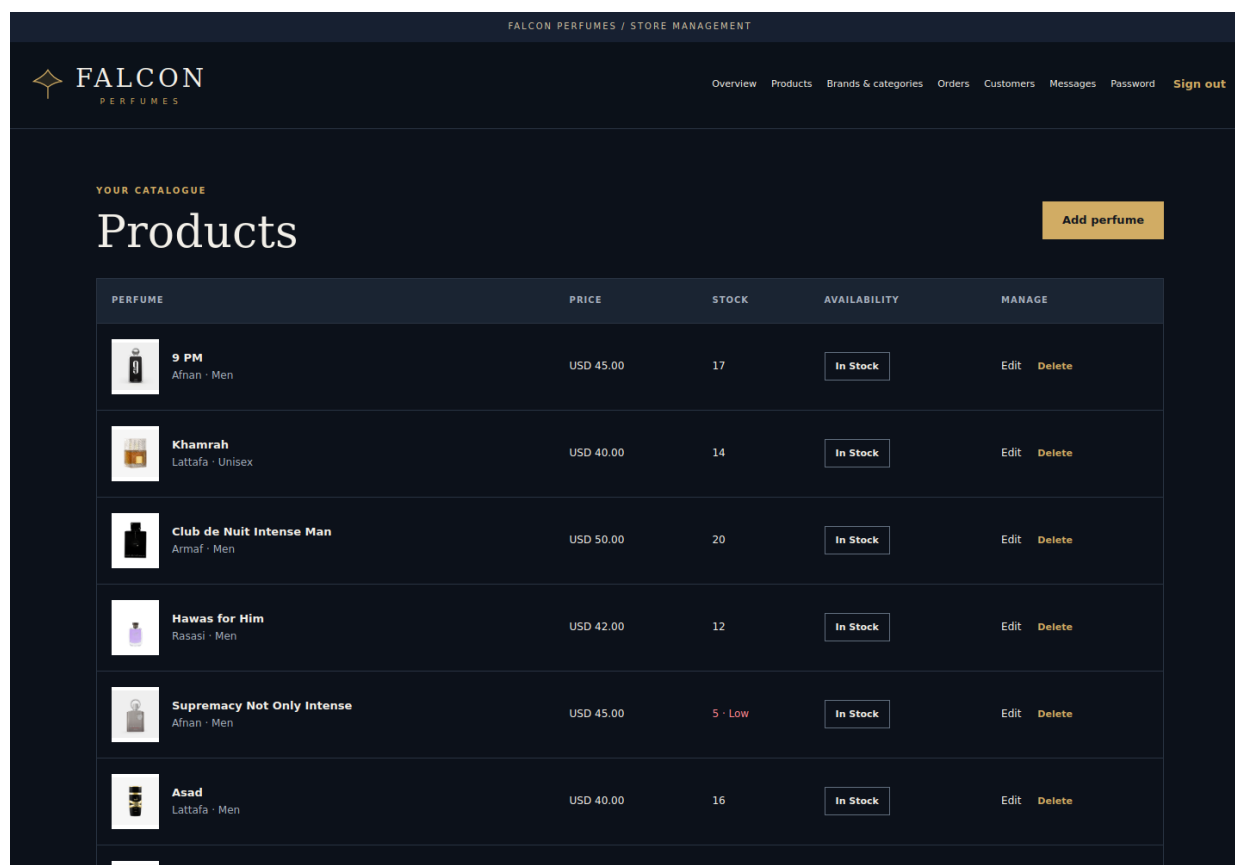
The catalogue contains eight identified fragrances with 2-3 locally bundled images each. Product pages display the brand, precise name, concentration, size, price, stock availability, description and top/heart/base notes. Selecting a thumbnail changes the main image; the image link still works when JavaScript is unavailable.

Correction	Result
Generic stock images	Replaced with product-matched bottle/packaging photos
Ambiguous product names	Specific Club de Nuit Intense Man, Supremacy Not Only Intense and Jean Lowe Immortel variants
Categories	Khamrah is unisex; Asad is in Men
Uncertain RIFFS Iconic listing	Replaced with identified Lattafa Yara; Riiffs remains a managed brand

The collection supports search suggestions, collapsible filters, price sorting and pagination. Related products below each detail page come from the same brand. Prices are editable demonstration values inherited from the frontend, not current market quotations.

06 / Implementation

Admin catalogue and availability management



The screenshot shows the 'FALCON PERFUMES / STORE MANAGEMENT' admin interface. The top navigation bar includes links for Overview, Products, Brands & categories, Orders, Customers, Messages, Password, and Sign out. The main section is titled 'YOUR CATALOGUE Products' and features an 'Add perfume' button. Below this is a table with columns: PERFUME, PRICE, STOCK, AVAILABILITY, and MANAGE. The table lists six products with their respective details and management options.

PERFUME	PRICE	STOCK	AVAILABILITY	MANAGE
 9 PM Afnan · Men	USD 45.00	17	In Stock	Edit Delete
 Khamrah Lattafa · Unisex	USD 40.00	14	In Stock	Edit Delete
 Club de Nuit Intense Man Armaf · Men	USD 50.00	20	In Stock	Edit Delete
 Hawas for Him Rasasi · Men	USD 42.00	12	In Stock	Edit Delete
 Supremacy Not Only Intense Afnan · Men	USD 45.00	5 · Low	In Stock	Edit Delete
 Asad Lattafa · Men	USD 40.00	16	In Stock	Edit Delete

Figure 3. Protected product management list, with prices, stock and availability controls.

Every protected admin page loads the staff-session guard before rendering. The temporary admin password must be changed at first login. Staff can add or edit perfumes, manage brands and categories, upload multiple photos, choose the cover image, remove gallery images and archive or restore a product.

Uploads accept JPEG, PNG and WebP after MIME and image-dimension checks. Each photo is limited to 5 MB and 20 megapixels, and a product must retain 2-8 images. Uploaded filenames are random; executable extensions are blocked in the upload directory.

The In Stock / Sold Out control updates a database flag. Zero stock also makes a product unavailable. The storefront shows a red SOLD OUT badge and a disabled grey purchase button. PHP repeats availability checks for cart updates and order placement, so editing browser HTML cannot purchase an unavailable product.

Low-stock warnings appear at five units or fewer. A brand or category linked to a product cannot be deleted until its references are removed or reassigned.

07 / Implementation

Orders, stock transactions and dashboard totals

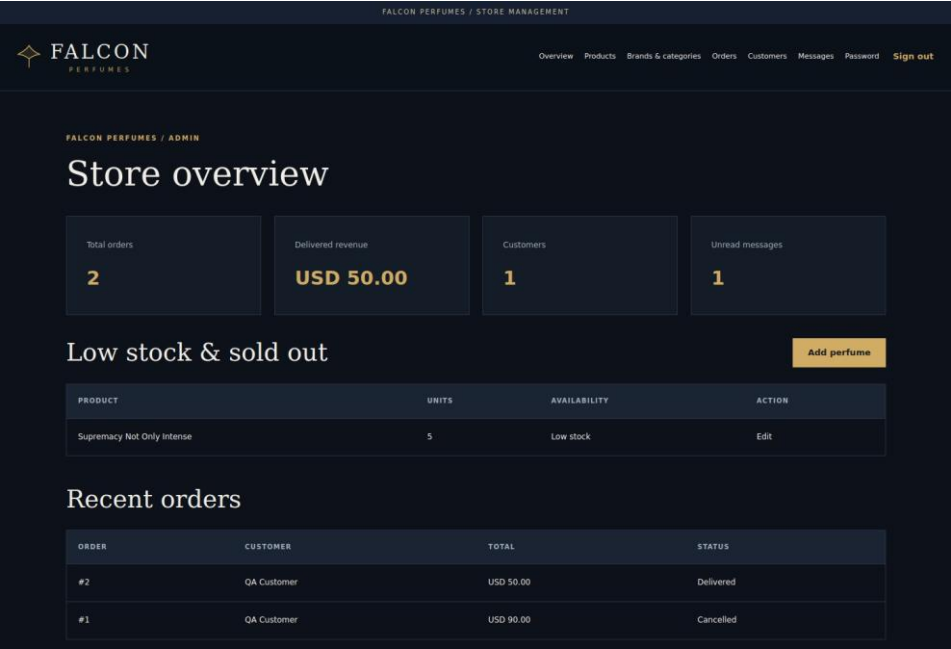


Figure 4. Admin overview with demonstration order totals and low-stock warnings.

The cart stores product IDs and quantities in the customer session. Prices and totals are loaded from the database. Checkout requires login and a delivery name, phone and address. The server calculates totals in integer cents, applies USD 5 delivery below USD 50, and creates a cash-on-delivery order.

Order creation runs inside an InnoDB transaction. Product rows are locked in a consistent ID order, stock is rechecked and decremented, and the order plus items are saved together. A unique checkout token and redirect prevent accidental duplicate submissions.

Current status	Allowed next status
Pending	Confirmed or Cancelled
Confirmed	Shipped or Cancelled
Shipped	Delivered
Delivered / Cancelled	No further change

Cancellation restores each quantity once, under an order-row lock. The dashboard counts total orders, customers, unread messages and delivered revenue. Revenue includes shipping on Delivered cash-on-delivery orders; no online payment settlement is claimed.

08 / Implementation

Contact form and customer service

The screenshot shows the Falcon Perfumes website interface. At the top, a dark header contains the Falcon Perfumes logo on the left and navigation links (Home, New Arrivals, Men, Women, Unisex, Brands, Contact, Login, Bag) on the right. A banner below the header reads "WE ARE HERE TO HELP" and "Let's talk fragrance." The main content area is divided into two columns. The left column features a "Send us a message" form with fields for "Your name", "Email address", and "Message", followed by a "Send message" button. The right column contains promotional text: "A little guidance. A lasting impression." and "Have a question about a fragrance, your order or delivery? Send your query and our team will review it." Below this is a box titled "Order enquiries" with the text "Include your order number so we can help you faster." and a link "View your orders →".

Figure 5. Contact form connected to the messages table.

The contact form collects name, email and message. Client-side and server-side checks enforce valid lengths and email format. A valid CSRF-protected request inserts a prepared statement into messages, then redirects with a success confirmation. A short session-based cooldown limits repeated submissions.

The admin Messages page lists sender details, timestamp and query, with read/unread controls. Message text is escaped before rendering; a test containing script markup appeared as text and did not execute. This implementation stores enquiries but does not send emails.

The Customers page shows registered users and their order histories. Staff can disable and re-enable accounts. Public contact details from the original mockup were not treated as verified business information, so the finished contact page focuses on the working enquiry form and order assistance.

09 / Validation

Challenges, solutions and test evidence

Challenge	Solution implemented
Missing frontend assets	Recreated CSS/JS locally; shared PHP templates retain Falcon branding and the four page roles.
Mismatched bottle imagery	Matched each precise fragrance variant with downloaded product photos and recorded source URLs.
Trusting browser state	Repeated validation, availability and price calculations in PHP; customer input cannot set an order total.
Stock consistency	Transactions and row locks protect stock deduction; permitted transitions prevent repeated cancellation restocking.
Customer/admin separation	Independent tables and sessions, centralized guards and forced first-login password replacement.
Template variable collision	Visual QA exposed an admin form rendering issue; navigation variables were isolated and field rendering rechecked.
Local setup portability	No external frontend assets; configurable subfolder/base URL; tested MySQL-compatible schema and documented XAMPP setup.

Completed verification

Real HTTP and browser tests exercised customer registration and duplicate rejection, login session rotation, admin isolation, missing-CSRF rejection, SQL-injection login rejection, escaped stored text, cart pricing, order creation, duplicate checkout, stock changes, cancellation, account disabling, product CRUD, image upload validation and taxonomy constraints.

Chromium checks verified gallery switching, mobile menu behavior, loaded product images, no JavaScript errors and no horizontal overflow on six customer pages at 390 px. PHP source files passed syntax checks. Read-only CLI smoke checks validate tables, gallery files and bcrypt hashes. The detailed check list is in tests/TEST-RESULTS.md.

Test environment: PHP 8.3.6, MariaDB 10.11.14 and Chromium 131 on Linux. The XAMPP graphical control panel was not run here; the student group must complete the documented XAMPP run before submission. Screenshots use temporary demonstration accounts and orders, which are excluded from the shipped database.

10 / Setup instructions

Run locally in XAMPP and export with phpMyAdmin

Install and import

1. Install XAMPP with PHP 8.2+ and start Apache and MySQL. Extract the ZIP and place the inner falcon-perfumes folder in C:\xampp\htdocs\ . On macOS use /Applications/XAMPP/xamppfiles/htdocs/.
2. Open <http://localhost/phpmyadmin>. Select Import, choose database.sql and run the SQL import. It creates falcon_perfumes with the tables, catalogue and temporary admin hash. Import into a fresh database; the SQL intentionally avoids dropping existing tables.
3. Check includes/config.php. Local defaults: host 127.0.0.1, port 3306, database falcon_perfumes, username root, blank password, base URL /falcon-perfumes. For changes, copy config.local.example.php to config.local.php and edit that file.

Open and demonstrate

4. Visit <http://localhost/falcon-perfumes/>. Register a customer through auth/register.php, sign in, open a perfume, add it to the bag and place a cash-on-delivery order.
5. Open <http://localhost/falcon-perfumes/admin/login.php>. Read the randomly generated temporary username/password in ADMIN-LOGIN.txt. Change the password when prompted. Edit a product, inspect orders and review a contact message.
6. For uploads, enable pdo_mysql, mbstring and fileinfo, and make images/uploads writable. For eight-image batches use upload_max_filesize=5M and post_max_size=45M in php.ini, then restart Apache. Do not open PHP pages by double-clicking them.

Export and submit

7. The included database.sql is a clean MariaDB export generated after testing. To meet the assignment wording exactly, export your local database through phpMyAdmin: select falcon_perfumes, choose Export, Quick, SQL, then Export. Put the resulting database.sql in your submission copy.
8. Create or update the project GitHub repository and submit the source ZIP, this report PDF and the repository URL on LMS. Confirm the official deadline in LMS: the supplied brief lists Sunday, 7 September 2026, although that calendar date is a Monday.

Troubleshooting

If connection fails, confirm MySQL is running, the database is imported and credentials/port match. If assets fail, correct base_url or remove an extra nested folder. If uploads fail, check PHP size limits and folder permissions. README.md includes CLI smoke checks and the local admin password-reset command.

11 / Individual contribution

Responsibilities and shared project work

The project was completed by a two-member group for ICT 1209 - Web Technologies. The work was divided between the customer-facing frontend and the PHP/MySQL backend, with both members contributing to testing, documentation and the final report.

Member / index number	Individual contribution
P. Mubarak 2754	Frontend development and product content: adapted the Home, New Arrivals, Brands and Contact pages; worked on the responsive navy-and-gold layout, navigation, search/filter interface and product gallery; reviewed perfume names, descriptions and image matching. Shared responsibility for integration testing, documentation and report preparation.
M. F. M. Fazlulla 2722	Backend development: worked on the PHP/MySQL database integration, customer registration and login, separate admin authentication, product and image management, stock controls, order processing, customer management and message storage. Shared responsibility for integration testing, documentation and report preparation.

Shared responsibilities

Both members worked on reviewing the integration between the frontend and backend, checking customer and administrator workflows, identifying issues, preparing setup instructions and compiling the project report.

Submission deliverables

The submission consists of the complete source-code ZIP, database.sql, this project report PDF and a GitHub repository link supplied through LMS. The source package includes setup instructions, screenshots, product-image references and local verification checks.

Scope and limitations

This is an academic local-store implementation with cash on delivery. It does not implement online card payments, automatic email delivery, password-reset emails, two-factor authentication, refunds, a tax engine or courier integration. Admin lists are designed for a small project dataset; the public catalogue is paginated.

For public deployment, sample values should be replaced, image-use rights verified, and a dedicated database user and HTTPS configured. Credentials should be rotated and setup, report and SQL files restricted. Product-image copyrights remain with their owners.

12 / References

Technical documentation and image attribution

Documentation and product references for the implementation and catalogue. Product pages and image URLs were retrieved on 27 September 2026. Short descriptions are original summaries. Local filename-to-source mappings appear in docs/image-sources.json.

Reference	URL
PHP: password_hash	https://www.php.net/manual/en/function.password-hash.php
PHP: password_verify	https://www.php.net/manual/en/function.password-verify.php
PHP: session regeneration	https://www.php.net/manual/en/function.session-regenerate-id.php
PHP: PDO prepared statements	https://www.php.net/manual/en/pdo.prepared-statements.php
PHP: file uploads	https://www.php.net/manual/en/features.file-upload.php
MySQL: InnoDB locking reads	https://dev.mysql.com/doc/refman/8.0/en/innodb-locking-reads.html
XAMPP	https://www.apachefriends.org/
phpMyAdmin import/export	https://docs.phpmyadmin.net/en/latest/import_export.html
Afnan 9 PM	https://afnan.com/products/9-pm
Afnan Supremacy Not Only Intense	https://afnan.com/products/supremacy-not-only-intense
Lattafa Khamrah / Asad / Yara	https://www.lattafa-usa.com/products/khamrah ; /products/asad ; /products/yara
Armaf Club de Nuit Intense Man EDT	https://armafperfume.com/products/armaf-club-de-nuit-intense-eau-de-toilette-for-man-105ml
Rasasi Hawas product listing	https://feel22.com/products/rasasi-hawas-men-eau-de-parfum
Jean Lowe Immortel product listing	https://palette21.com/products/jean-low-immortel

Frontend source: the four user-supplied Falcon Perfumes HTML files, retained in docs/original-frontend/. Product photos remain third-party imagery. The project does not claim authorship of those photographs or a commercial reuse license.